

Fruit Store Checkout

Fruit detection, counting, and automated checkout support for supermarket-style environments.

Team 16 | MIE1517 Final Presentation

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Problem: Why Fruit Store Checkout?

What problem are we solving?

- Automatically identify, count fruits in real checkout environment

Why is it important?

- No barcode
- PLU entry time-consuming
- Cluttered, stacked, and partially occluded real cases

Why were we interested?

- Practical computer vision problem
- Deep learning in a real application
- From model training to deployment thinking



Image source: <https://locsoftware.com/canadian-grocery-pos/>

Manual PLU entry takes 3x longer than barcode scanning.

Live Demo: Realistic Detection

Team 16 Fruit Store Checkout App

Upload an image or video to detect fruits, count them, and generate a billing summary :)

Image Detection Video Detection

Upload Video



Detection Result

he count:

None

Unique tracked count:

None

Confidence Threshold

0.1

0.55

0.9

Video Summary and Billing

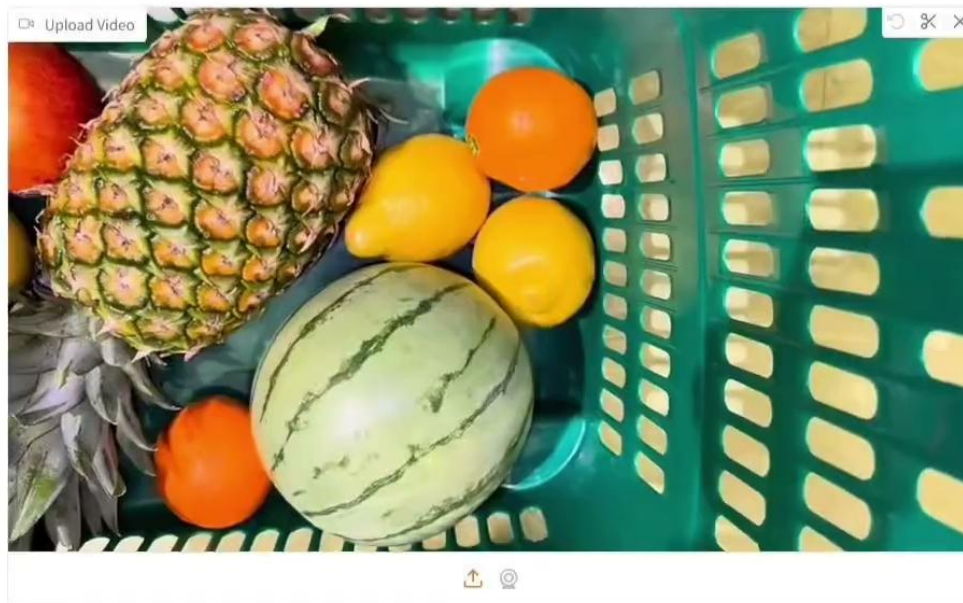
Live Demo: Realistic Detection

Team 16 Fruit Store Checkout App

Upload an image or video to detect fruits, count them, and generate a billing summary :)

Image Detection

Video Detection



Confidence Threshold

0.1

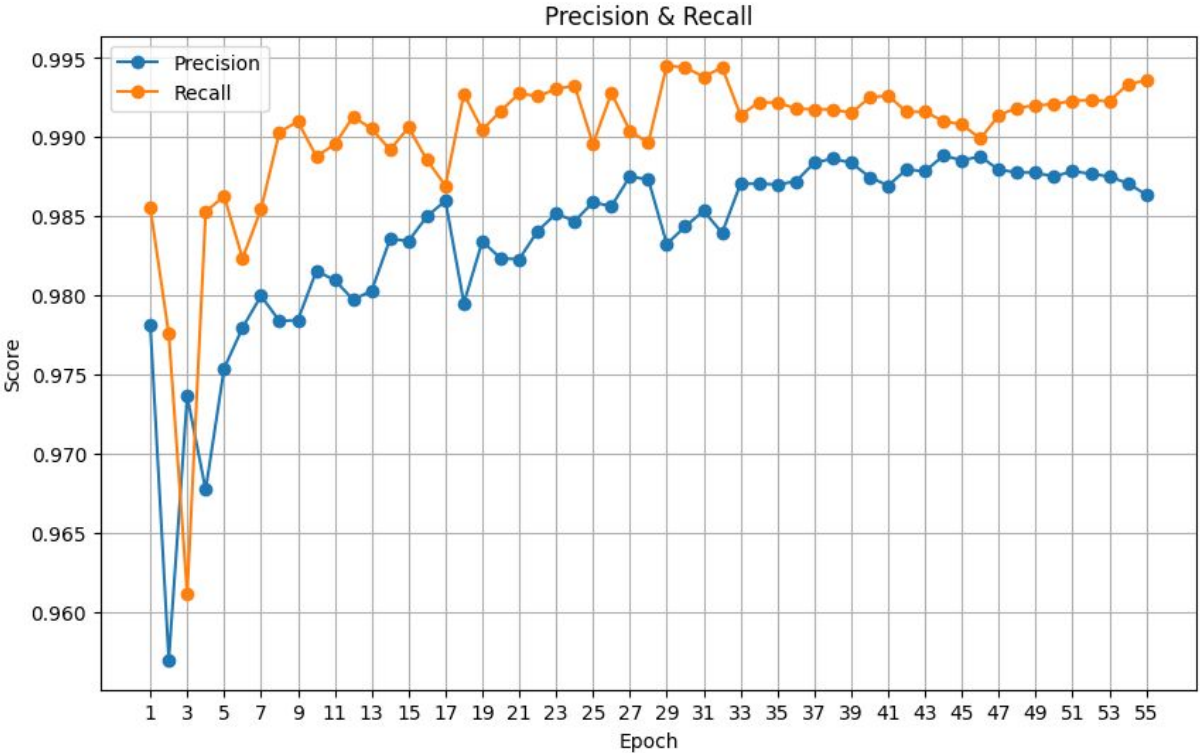
0.55

0.9

Results: Quantitative & Qualitative Analysis

Quantitative:

Metric Type	Model Result
Precision	0.985+
Recall	0.990+
mAP @ 0.50	0.995
mAP @ 0.5:0.95	0.947
Inference Speed	~27 ms/img



Results: Quantitative & Qualitative Analysis

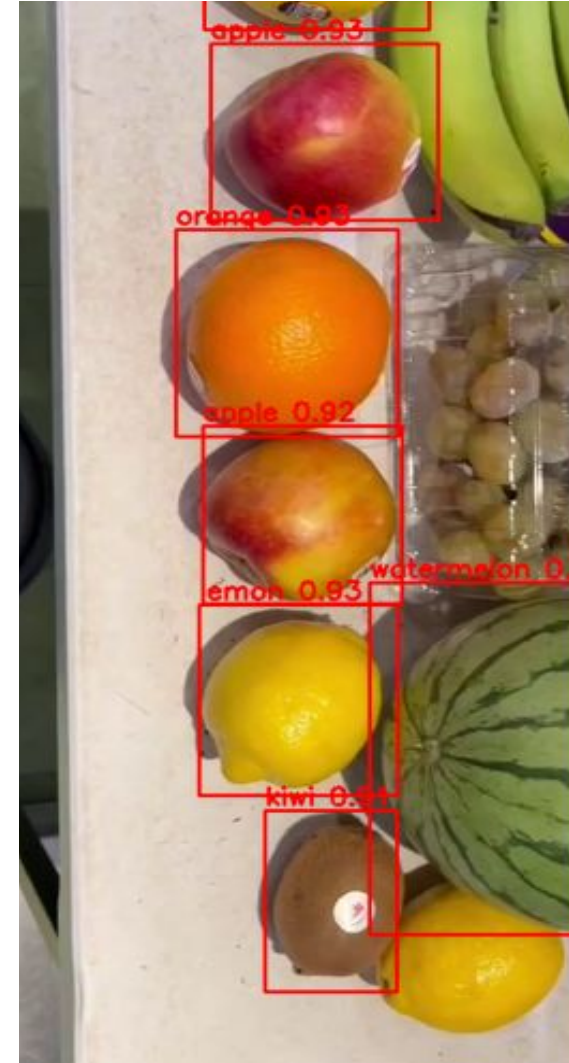
Qualitative:

Strengths:

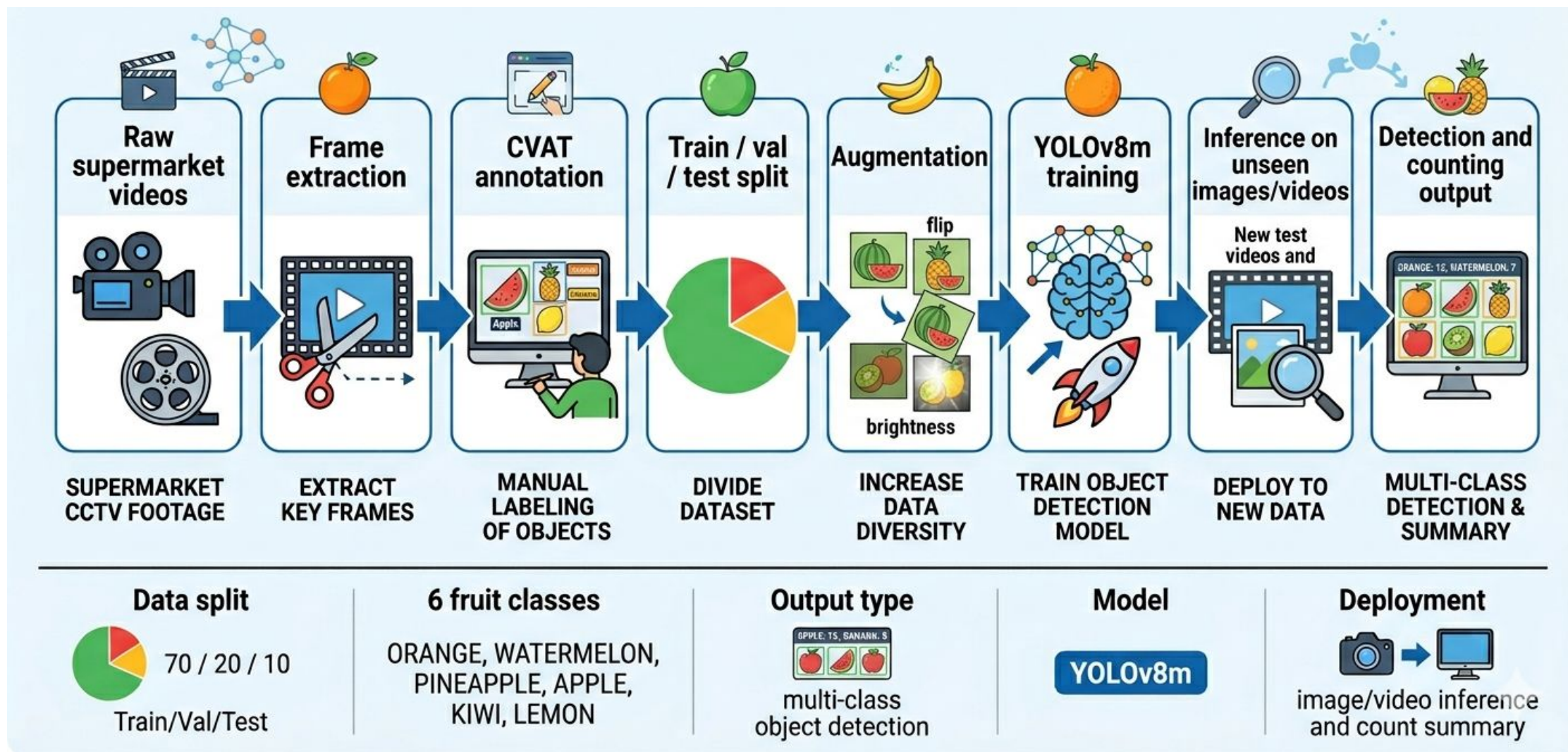
- Strong detection in cluttered scenes
- Good localization for most fruits
- Stable multi-object recognition

Failure cases:

- Overlap
- Motion blur
- Small or partially hidden fruits



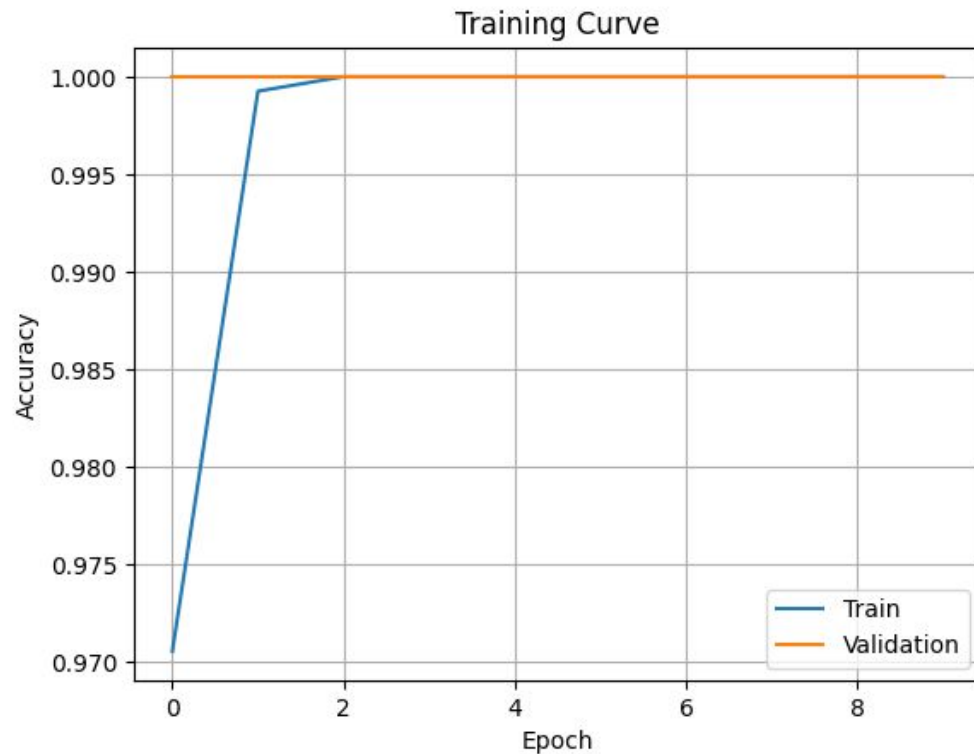
End-to-End Pipeline



Discussion: What We Learned

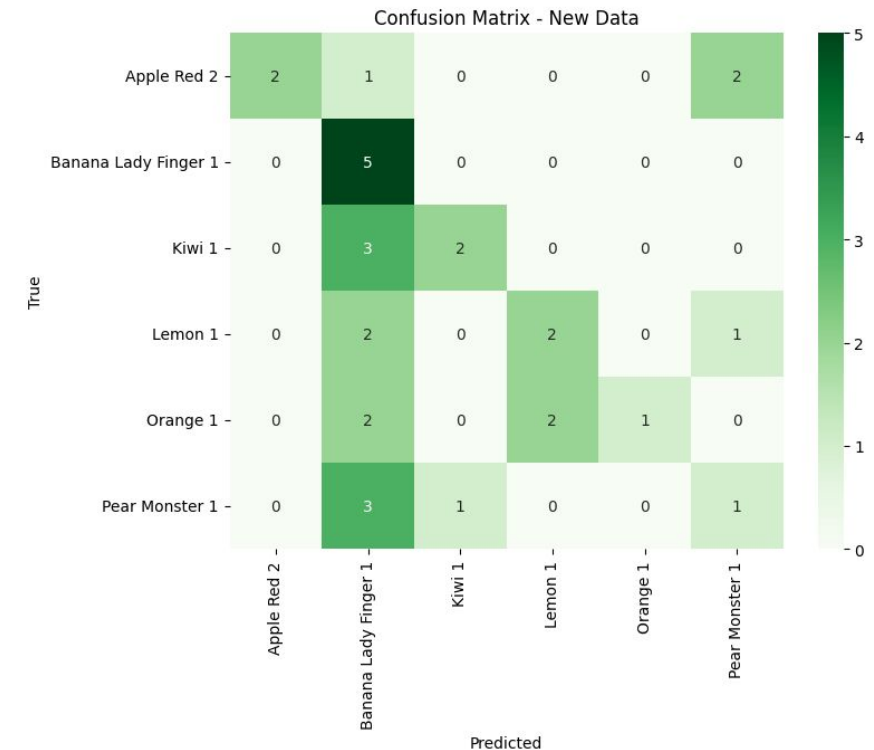
Base Model Limitation

- ResNet18 hit 100% on clean data but failed in real-world case. We learned that high accuracy on static datasets hides domain rupture.



Transition to final model

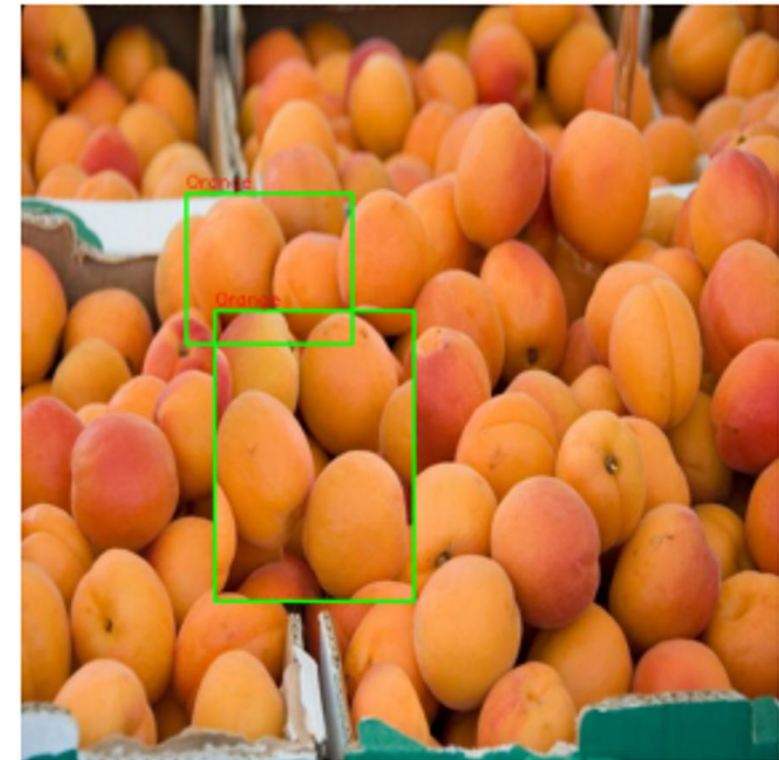
- Classification can't handle multiple fruits or counting. Transitioning to YOLO v8 was essential for a practical checkout system.



Discussion: What We Learned

YOLO v8m

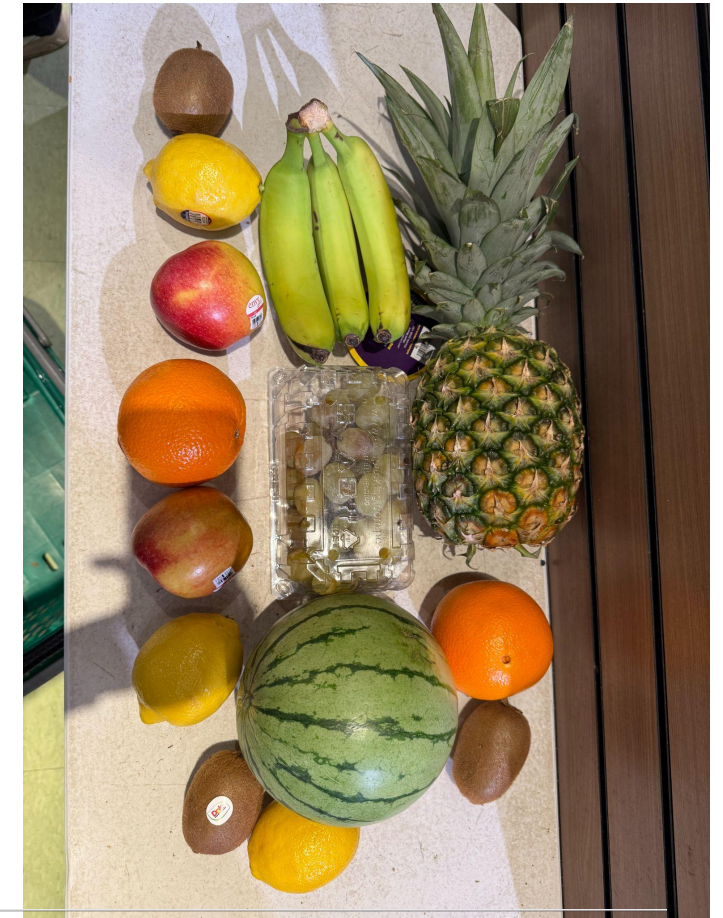
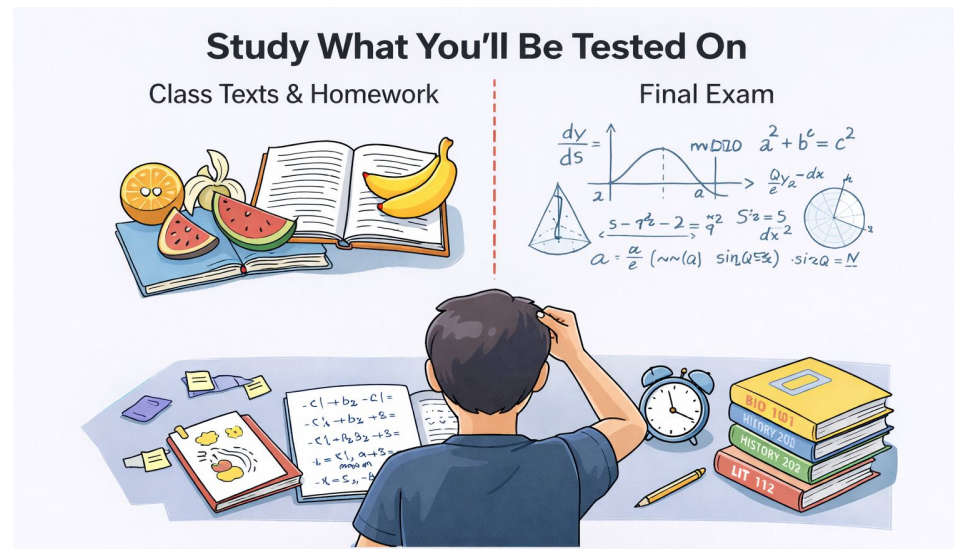
- Data from the internet may be some kinds of "dirty", because of inappropriate, incorrect, unbalanced.



Discussion: What We Learned

YOLO v8m

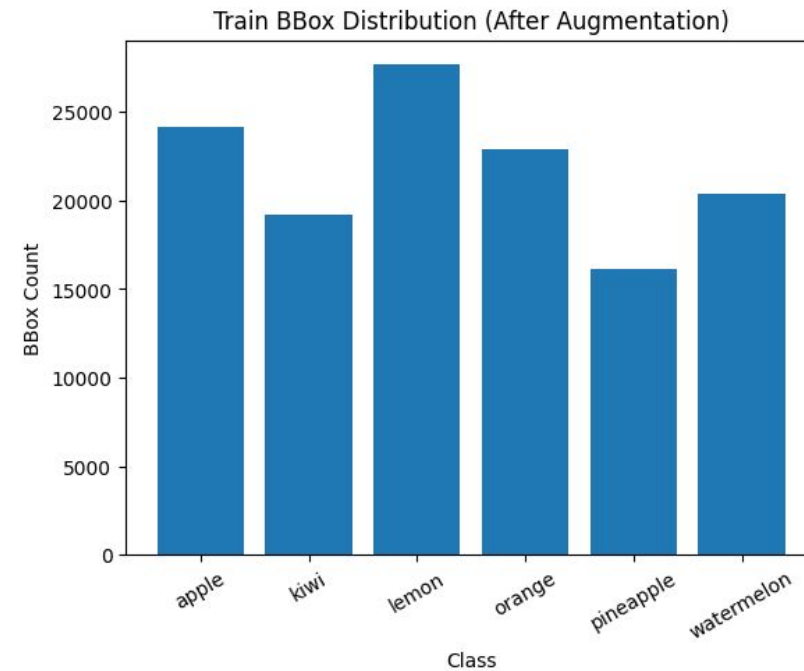
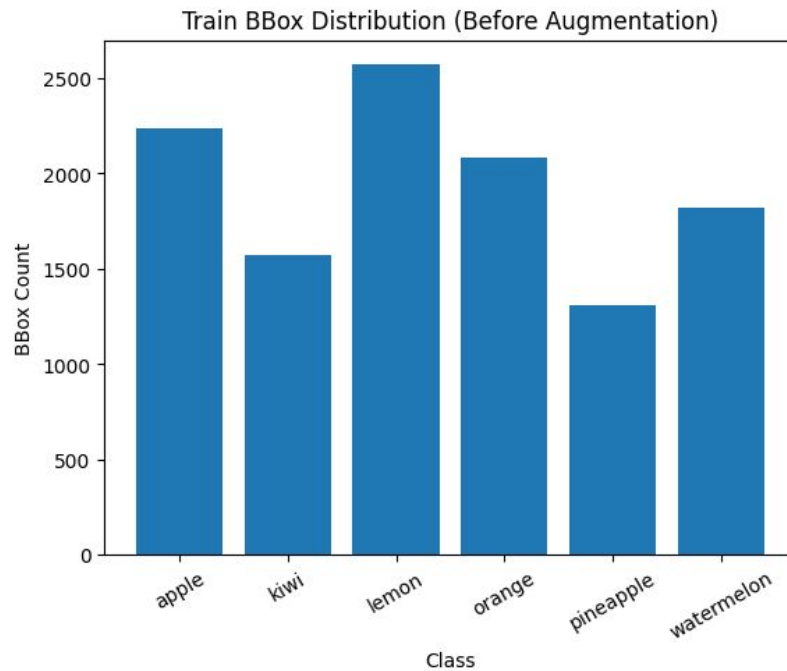
- Training must be conducted using scenarios similar to test. "Learn what you will be tested on."



Discussion: What We Learned

YOLO v8m

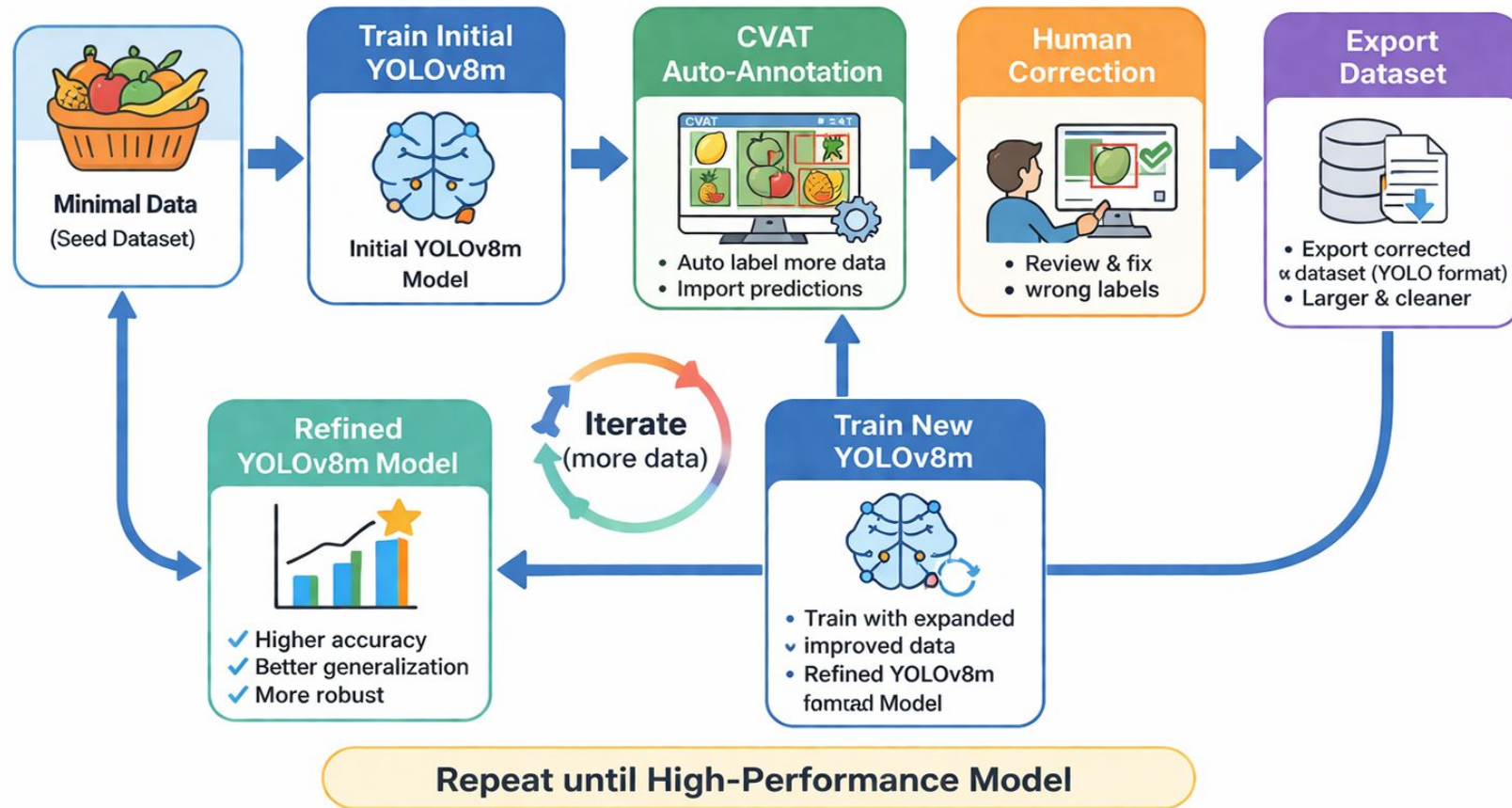
- Data augmentation is very important for genralization.
- Data augmentation needs to consider the balance of the training set compatibly.



Future Work

Self-Accelerating Model Training System

Based on Semi-Automatic Annotation with Human Correction



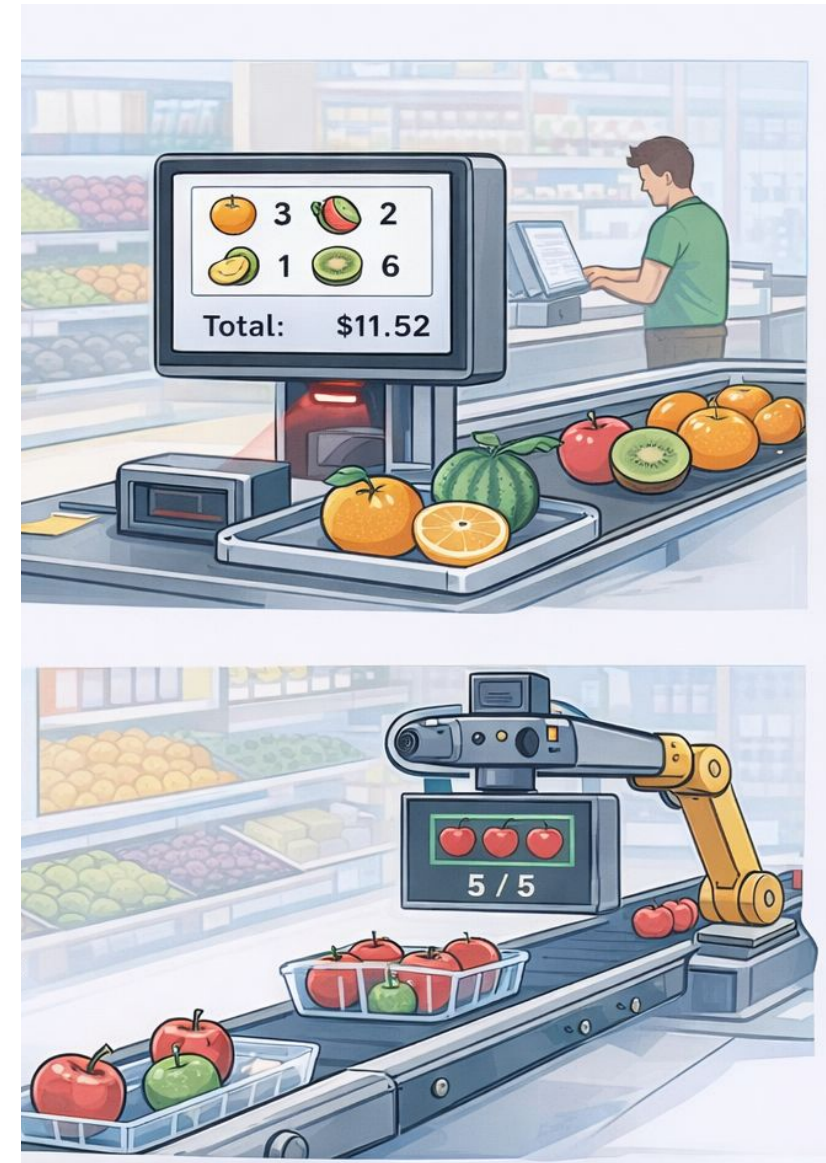
Conclusion

Practical lessons

- Data itself is the most crucial thing for deep learning models.
"Learn what you will be tested on."
- Data augmentation and balance is indispensable.
- A good model will not overfit quickly.

Future work

- Expand from 6 fruits to more categories.
- Improve generalization across more scenes.
- Solve tracking problems for auto-annotation.
- Deploy the model on checkouts, streamline, etc.



Reference

1. Fruits-360 Dataset (source: <https://www.kaggle.com/datasets/moltean/fruits>). Used for training of ResNet18 model. Name: fruit_6.
2. Fruit Detection Dataset (source: <https://www.kaggle.com/datasets/lakshaytyagi01/fruit-detection/data>). This set is annotated in YOLOv8 format for training of YOLOv8s model. Name: fruit_yolo.
3. MVTec D2S (source: <https://datasetninja.com/mvtec-d2s>). Used for later formal training with YOLOv8m to adapt fruits, vegetables and packages. Name: mdfull.
4. Generalization test set (source: from the internet and a few phone taken photos; all in .jpg format and 100x100 size). Worked on both ResNet18 and YOLOv8s. Name: fruit_6_new.

Welcome to have a try!

